AGENTBIO — SAVED HYPOTHESIS REPORT

Chronic-disease indications predict lower repurposing success than non-chronic indications.

AUDIT NUMBERS — FROM THE REGISTRY AT SAVE TIME

FRAME	TEST	OR	95% CI	N	DISCOVERY RAW P	FDR Q	CONFIRM RAW P
narrow	fisher_exact	0.755	[0.284, 1.7]	2913	5.89e-1	8.42e-1	–
broad	fisher_exact	1.47	[1.11, 1.92]	4203	5.90e-3	1.90e-2	1.21e-4

CONFOUND	UNADJ. OR	ADJ. OR	ADJ. 95% CI	ADJ. P	SURVIVES?
prior-repurposing saturation of chronic indications	1.47	2.20	[1.6479, 2.9331]	0.00e+0	survives
established-product incumbency in chronic markets	1.47	1.49	[1.1275, 1.9611]	4.96e-3	survives
CNS/psychiatric indication difficulty	1.47	1.31	[0.9849, 1.7376]	6.36e-2	survives

NARRATIVE — CLAUDE OPUS 4.8

Repurposing-Association Report: Chronic-Disease Indications and Repurposing Success

Hypothesis ID: run-09c577de-H28

Hypothesis

The hypothesis, stated verbatim, is: *"Chronic-disease indications predict lower repurposing success than non-chronic indications."* It was proposed by the model **Sol**, drawing on the source domain **"cycle-life dominance in rechargeable batteries under repeated duty cycles"** — an analogical transfer motivated by the idea that repeated dosing amplifies low-frequency toxicity, adherence, and exposure defects that may remain tolerable in short-course treatment.

Prior-literature tag (web-search, informational): UNCLEAR.

Bisociative provenance

This finding comes from run **run-09c577de**. Both generators contributed candidate source domains:

Opus proposed:

- impedance matching in transmission-line power transfer
- loss aversion / endowment effect in behavioral decision theory
- quorum sensing thresholds in bacterial gene regulation
- stigmergy / pheromone-trail reinforcement in ant foraging
- vestibular sensory reweighting under conflicting cues (multisensory integration)

Sol proposed:

- cycle-life dominance in rechargeable batteries under repeated duty cycles
- impedance matching by transformer networks in radio-frequency circuits
- kinetic proofreading under near-cognate T-cell receptor recognition
- mode matching at refractive-index discontinuities in optical waveguides

Lead reviewer consolidation (from lead_decisions): The `lead_decisions` array records per-framing effect summaries and gating outcomes rather than explicit READY / NEEDS_ENRICHMENT / DISCARDED status labels. For the specific hypothesis under this report ("Chronic-disease indications predict lower repurposing success than non-chronic indications," proposed by Sol from the cycle-life domain), the lead decisions record two computed framings: one with **OR=0.755, CI[0.284, 1.7], n=2913**, and one with **OR=1.47, CI[1.11, 1.92], n=4203**, each accompanied by the mechanistic note about repeated dosing amplifying low-frequency toxicity, adherence, and exposure defects. Because the provided records do not carry an explicit READY/NEEDS_ENRICHMENT/DISCARDED token for this or the other hypotheses, I cannot assign those categorical labels without guessing; I therefore report only what is present — the effect estimates and the recorded rationale text — and note that many sibling hypotheses in the same run were instead marked "not tested" (e.g., "perfect separation within a moderator level," "base is constant within a moderator level") or "SKIPPED (duplicate)."

How this was actually tested

The literal computable proxy applied to the dataset was:

indication name (lowercased) contains any of: [chronic, diabetes, hypertension, arthritis, asthma, epilepsy, depression, osteoporosis]

This keyword-matching proxy — not the broader biological claim about repeated dosing and cumulative toxicity — is what the statistics below actually measured.

Statistical evidence underneath the p-value

Two outcome framings are present in FACTS. Both used **Fisher's exact test**.

FRAMING	TEST	ODDS RATIO	95% CI	N	DISCOVERY RAW P	FDR Q
narrow	fisher_exact	0.755	[0.284, 1.7]	2913	0.5888224751255567	0.8422717482848417
broad	fisher_exact	1.47	[1.11, 1.92]	4203	0.005903204211165828	0.01900093855469001

> **Narrow framing:** The odds ratio of 0.755 (below 1) would nominally indicate lower odds of the coded outcome for chronic-keyword indications, but the confidence interval [0.284, 1.7] crosses 1 and the effect is not statistically distinguishable from no effect (raw p = 0.5888224751255567).
>
> **Broad framing:** The odds ratio of 1.47 (above 1) indicates a moderately *higher* odds of the coded outcome for chronic-keyword indications, with a confidence interval [1.11, 1.92] entirely above 1 — a direction opposite to the "lower success" wording of the hypothesis.

Discovery vs. confirmation

FRAMING	DISCOVERY RAW P	FDR Q	DISCOVERY PASS	CONFIRMATION RAW P	CONFIRMATION PASS
narrow	0.5888224751255567	0.8422717482848417	false	null	false
broad	0.005903204211165828	0.01900093855469001	true	0.00012116168851134307	true

Only the **broad** framing passed both the FDR-corrected discovery gate and the independent holdout-confirmation test. Passing confirmation means the effect reproduced on data never used during discovery (confirmation raw p = 0.00012116168851134307). The narrow framing did not pass discovery and has a **null** confirmation p-value (no confirmation result recorded); it did not pass confirmation.

Confound checks

The `confound_check` object (status: **completed**; note: "candidate explanations checked — not the confirmed mechanism") lists three specific alternative explanations. Each was computable. Unadjusted reference values are OR = 1.4675, p = 0.005903, n = 4203.

1. Prior-repurposing saturation of chronic indications

> *Rationale:* Chronic-disease indications like diabetes and hypertension are large commercial targets already repurposing-attempted many times, so remaining candidates are picked-over and enriched for prior failures; a high prior_repurposing_count, not the chronic mechanism, might then drive the effect.
>
> *Computable:* Yes.
>
> *Unadjusted:* OR = 1.4675, p = 0.005903.
>
> *Adjusted:* OR = 2.1985, 95% CI [1.6479, 2.9331], p = 0.0.
>
> *Survived adjustment:* **Yes** ("effect survives adjustment").

2. Established-product incumbency in chronic markets

Rationale: Chronic indications are dominated by entrenched established products, so a repurposing entrant must clear a high standard-of-care bar independent of dosing toxicity; "established" status may confound the chronic keyword because chronic-disease drugs are disproportionately established.

Computable: Yes.

Unadjusted: OR = 1.4675, p = 0.005903.

Adjusted: OR = 1.487, 95% CI [1.1275, 1.9611], p = 0.004957.

Survived adjustment: **Yes** ("effect survives adjustment").

3. CNS/psychiatric indication difficulty

Rationale: The keyword list includes "depression" and "epilepsy" (CNS indications) with high clinical-trial attrition from placebo response and endpoint subjectivity rather than chronic dosing; the apparent chronic effect could be a repackaged CNS-difficulty effect from a keyword subset.

Computable: Yes.

Unadjusted: OR = 1.4675, p = 0.005903.

Adjusted: OR = 1.3082, 95% CI [0.9849, 1.7376], p = 0.063625.

Survived adjustment: **Yes** per the recorded flag ("effect survives adjustment"). Note, for the record: after this adjustment the confidence interval [0.9849, 1.7376] crosses 1 and the adjusted p (0.063625) exceeds the 0.05 threshold, even though the `survives_adjustment` field is marked true.

Verdict: The effect survived adjustment for **every** computable confound according to the recorded `survives_adjustment` flags — prior-repurposing saturation, established-product incumbency, and CNS/psychiatric difficulty were all checked and all returned survives_adjustment = true. The one caveat that must be flagged transparently is the CNS/psychiatric adjustment: while marked as surviving, its adjusted 95% CI [0.9849, 1.7376] crosses 1 and its adjusted p-value (0.063625) is above 0.05, indicating the effect is materially attenuated once CNS content is accounted for.

Bottom line

In this curated dataset, a keyword-based coding of "chronic-disease" indications is statistically associated with the repurposing-success outcome in the broad framing (OR = 1.47, 95% CI [1.11, 1.92], n = 4203), surviving FDR correction, independent holdout confirmation, and adjustment for three named confounds — though notably the *direction* (OR above 1) is opposite to the hypothesis's stated prediction of "lower" success, and the narrow framing showed no significant effect. This is a statistical association measured through a lowercased keyword-matching proxy, not evidence of causation, biological mechanism, or clinical relevance. The strongest remaining caveat is that the CNS/psychiatric-difficulty adjustment leaves the effect substantially attenuated (adjusted OR = 1.3082, CI [0.9849, 1.7376], p = 0.063625, crossing 1), suggesting a meaningful portion of the association may be attributable to CNS-indication content within the keyword list.